

CHANDANI DEVI

QC / Analytical Chemist | Water & Environmental Testing | NABL / ISO 17025 | UV-Vis, AAS, Wet Chemistry | M.Sc Chemistry

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Delhi

SUMMARY

M.Sc Chemistry and QC / Analytical Chemist in a NABL-accredited (ISO/IEC 17025) environmental and water testing laboratory. Skilled in drinking-water and wastewater/effluent analysis — pH, TDS, hardness, alkalinity, chloride, sulphate, fluoride, nitrate, COD, BOD and heavy metals — as per IS 3025, APHA and IS 10500 methods, with additional experience in soil and ambient-air testing. Proficient in wet-chemistry titrations, UV-Vis spectrophotometry, sample digestion/preparation, instrument calibration and NABL-format reporting. Working knowledge of AAS, GC/GC-MS and HPLC.

EXPERIENCE

Chemical Analyst / QC Chemist

Eco Care Instruments Pvt. Ltd., Delhi

12/2025 - Present Delhi, India

- Water & Wastewater Testing :-**
- Performed physico-chemical analysis of drinking water, groundwater, raw water, and wastewater/effluent — pH, conductivity, TDS, turbidity, total hardness, alkalinity, chloride, sulphate, fluoride, nitrate, calcium, magnesium — as per IS 3025 / APHA / IS 10500 methods.
- Carried out titrimetric analysis (EDTA complexometric for hardness, argentometric for chloride, acid-base for alkalinity) and gravimetric determination of TDS / total solids.
- Conducted colorimetric analysis on UV-Vis spectrophotometer — Hexavalent Chromium (Cr VI), nitrate, fluoride, sulphate, ammonia, and phenol.
- Performed sample digestion and preparation for heavy-metal analysis (Fe, Mn, Zn, Cu, Cr, Pb, Cd, As, Hg, Ni) by AAS / spectrophotometric methods.
- Determined COD (potassium dichromate reflux) and BOD (5-day, 20°C incubation) for wastewater and effluent samples.
- Soil & Environmental Testing:-**
- Performed physico-chemical analysis of soil / sediment samples — pH, available nutrients, organic matter, and heavy metals — as per IS 2720 / APHA / EPA methods.
- Carried out ambient air quality monitoring — SO₂, NO₂, PM₁₀, PM_{2.5}, Ozone, CO — using high-volume sampling as per IS 5182 (CPCB NAAQS).
- Gained working exposure to stack-emission, noise-level, and illumination (Lux) monitoring as per IS 11255, IS 9989 / IS 7194, and IS 3646.
- Quality & Documentation:-**
- Prepared reagents and standards; performed daily instrument calibration and QC checks (blank, duplicate, spike) for ISO/IEC 17025 compliance, and documented NABL-format test reports for CPCB / SPCB and client requirements.

SKILLS



Analytical Techniques

Wet Chemistry, Titrimetric (Complexometric / Argentometric / Acid-Base), Gravimetric & Colorimetric Analysis, Sample Digestion & Preparation



Testing Domains

Drinking / Ground / Waste Water, Soil & Sediment, Ambient Air, Stack Emission, Noise, Illumination (Lux)



Instrument Proficiency

UV-Vis Spectrophotometer, pH Meter, Conductivity / TDS Meter, Turbidity Meter, COD Digestion Unit, BOD Incubator, DO Meter, Hot Air Oven, Muffle Furnace, Analytical Balance, Flame Photometer, High-Volume Air Sampler, Sound Level Meter, Lux Meter, AAS, GC / GC-MS, HPLC, ICP-MS



Standards & Quality

NABL, ISO/IEC 17025:2017, IS 3025, APHA, IS 10500, IS 2720, IS 5182, IS 11255, IS 3646, GLP, Instrument Calibration, Quality Control, Traceability, Chain of Custody

EDUCATION



Master of Science in Chemistry (M.S.C) - 66.4%

Mahatma Gandhi Kashi Vidyapith, Varanasi

05/2022 - 05/2024 Varanasi, Uttar Pradesh



Bachelor of Science in Chemistry (B.S.C) - 61.8%

Mahatma Gandhi Kashi Vidyapith, Varanasi

03/2019 - 05/2022 Varanasi, Uttar Pradesh



Intermediate (XII) - Science Stream - 53.8%

A.B. Vajpayee Intermediate College, Varanasi

05/2019 - Present Varanasi, Uttar Pradesh



High School (X) - Science Stream - 68.3%

A.B. Vajpayee Intermediate College, Varanasi

03/2016 - 05/2017 Varanasi, Uttar Pradesh

KEY ACHIEVEMENTS



Independent Laboratory Testing & Analysis

Independently handle daily physico-chemical testing of drinking-water and wastewater samples with accurate, reproducible results in a NABL-accredited lab.



Technical Expertise in Standard Analytical Methods

Built strong practical command of IS 3025 / APHA wet-chemistry methods and UV-Vis colorimetric analysis within a short span.



Quality Assurance & NABL Compliance

Maintain NABL / ISO 17025 documentation and QC discipline (blank, duplicate, spike) ensuring reliable, traceable test data.



Multi-Domain Testing Exposure

Gained hands-on exposure across water, wastewater, soil and ambient-air testing in a single NABL-accredited lab, building a broad analytical skill set within a short span.

INTERESTS

Analytical & Environmental Chemistry

Water Quality & Treatment | Laboratory Quality Systems (NABL / ISO 17025) | Continuous Learning in Analytical Instrumentation